

Shanshan Mao

Final-Year PhD Researcher
Multi-Agent Systems and Artificial Life
School of Computer Science
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Research Profile

Final-year PhD researcher in Computer Science at the University of Birmingham. Research focuses on dynamic multi-agent systems, artificial life, and complex adaptive systems, with particular emphasis on individual agency, emergent macro-level dynamics, evolutionary adaptation, and signal manipulation.

Education

University of Birmingham, UK
PhD in Computer Science

Mar 2021 – Aug 2026 (expected submission)

- Research focuses on emergence, stability, and collapse of hierarchy in multi-agent systems under bounded rationality and deceptive signalling.

University of Birmingham, UK
MSc in Advanced Computer Science

Sept 2019 – Sept 2020
Distinction

- Modules: Machine Learning, Neural Computation, Nature-Inspired Search, Intelligent Data Analysis.

Henan University, China
BSc in Computer Science and Technology

Sept 2014 – Jun 2018

- Final grade: 80.5/100

Publications

S. Mao, P. Tino

From Cooperation to Hierarchy: A Study of the Dynamics of Hierarchy Emergence in a Multi-Agent System.
EvoApplications 2026 (EvoStar), Springer. Accepted for publication.

S. Mao, P. Tino

When Deception Spreads: Transmissible Deception Destabilises Emergent Hierarchy in a Cooperative Multi-Agent System.
Artificial Life Conference (ALIFE 2026). Accepted.

Academic Presentations

Strategic Interaction and Emergent Hierarchy in Multi-Agent Systems

Lightning Talk, Game Theory in the Era of AI Workshop, 2026

How Hierarchies Emerge Without Leaders: Insights from Multi-Agent Systems

Presentation, Birmingham International Scholars Forum, Sub-Forum on Artificial Intelligence and Digitalization, 2026

How Does Hierarchical Structure Emerge in Decentralized Multi-Agent Systems

Presentation, Toulouse, France, Evostar Applications conference, 2026

Research Experience

PhD Researcher, University of Birmingham

Mar 2021 – Present

Emergence of Hierarchy in Multi-Agent Systems

- Developed an agent-based simulation framework for studying hierarchy emergence under local interaction and cooperation.

- Investigated the formation, persistence, and collapse of hierarchical structures under deceptive signalling and evolutionary adaptation.
- Applied trophic incoherence and network-based metrics to quantify structural organisation and disorder.
- Conducted large-scale simulation experiments and statistical analyses across multiple parameter settings and random seeds.

Master's Thesis, University of Birmingham

Nov 2019 – Jun 2020

Heuristic Optimisation for the Constrained Travelling Salesman Problem

- Investigated metaheuristic optimisation methods for constrained travelling salesman problems.
- Evaluated Genetic Algorithms, Tabu Search, and Simulated Annealing on benchmark optimisation tasks.

Teaching Experience

Postgraduate Teaching Assistant, University of Birmingham

Sept 2022 – Mar 2026

- Delivered tutorials and lab sessions in Data Structures, Databases, AI/ML, and Evolutionary Computation.
- Provided technical guidance and assessment support for large undergraduate cohorts.
- Assessed coursework, examinations, and programming assignments for large undergraduate cohorts.

Teaching Assistant, Machine Learning Summer School

Jul 2025

- Supported hands-on teaching of machine learning and deep learning frameworks (scikit-learn, PyTorch, TensorFlow).

Research Projects

AI Companion Robotics (Doctoral Innovation Programme)

Sept 2023 – Jun 2024

- Led development of a multi-modal AI companion prototype integrating speech, vision, and interaction modules.

User Interface Developer (Project)

2018

- Developed Python-based UI to interface with hardware simulation systems via API.

Skills

Programming: Python, C++, MATLAB, JavaScript

ML / Data: PyTorch, scikit-learn, NumPy, pandas, Matplotlib, Seaborn

Research Methods: Agent-based modelling, complex systems simulation, network analysis, statistical modelling, hypothesis testing, reinforcement learning environments

Tools: Git, Jupyter, VS Code, PyCharm, LaTeX

Awards

First Prize, Sustainable Future in the City Competition

Oct 2019